

Lab 8

Q1

Practice 1

```
1: ilovetrains28@cis106: ~  
15811 against accepting unsolicited donations from donors in such states who  
15812 approach us with offers to donate.  
15813  
15814 International donations are gratefully accepted, but we cannot make  
15815 any statements concerning tax treatment of donations received from  
15816 outside the United States. U.S. laws alone swamp our small staff.  
15817  
15818 Please check the Project Gutenberg web pages for current donation  
15819 methods and addresses. Donations are accepted in a number of other  
15820 ways including checks, online payments and credit card donations. To  
15821 donate, please visit: www.gutenberg.org/donate.  
15822  
15823 Section 5. General Information About Project Gutenberg™ electronic works  
15824  
15825 Professor Michael S. Hart was the originator of the Project  
15826 Gutenberg™ concept of a library of electronic works that could be  
15827 freely shared with anyone. For forty years, he produced and  
15828 distributed Project Gutenberg™ eBooks with only a loose network of  
15829 volunteer support.  
15830  
15831 Project Gutenberg™ eBooks are often created from several printed  
15832 editions, all of which are confirmed as not protected by copyright in  
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15837 Most people start at our website which has the main PG search  
15838 facility: www.gutenberg.org.  
15839  
15840 This website includes information about Project Gutenberg™,  
15841 including how to make donations to the Project Gutenberg Literary  
15842 Archive Foundation, how to help produce our new eBooks, and how to  
15843 subscribe to our email newsletter to hear about new eBooks.  
15844  
15845  
ilovetrains28@cis106:~$
```

Practice 2

```
1: ilovetrains28@cis106: ~  
Nutr#Honey Crunch calories = 120  
Nutri-Grain Almond-Raisin calories = 140  
Nutri-grain Wheat calories = 90  
Oatmeal Raisin Crisp calories = 130  
Post Nat. Raisin Bran calories = 120  
Product 19 calories = 100  
Puffed Rice calories = 50  
Puffed Wheat calories = 50  
Quaker Oat Squares calories = 100  
Quaker Oatmeal calories = 100  
Raisin Bran calories = 120  
Raisin Nut Bran calories = 100  
Raisin Squares calories = 90  
Rice Chex calories = 110  
Rice Krispies calories = 110  
Shredded Wheat calories = 80  
Shredded Wheat 'n'Bran calories = 90  
Shredded Wheat spoon size calories = 90  
Smacks calories = 110  
Special K calories = 110  
Strawberry Fruit Wheats calories = 90  
Total Corn Flakes calories = 110  
Total Raisin Bran calories = 140  
Total Whole Grain calories = 100  
Triples calories = 110  
Trix calories = 110  
Wheat Chex calories = 100  
Wheaties calories = 100  
Wheaties Honey Gold calories = 110  
ilovetrains28@cis106:~$
```

Practice 3

```
1/1 v + [ ] [ ]
Tilix: ilovetrains28@cis106: ~/Documents/sample_files/Csv
1: ilovetrains28@cis106: ~/Documents/sample_files/Csv
Volkswagen Dasher (diesel);43.4;4;90.00;48.00;2335.;23.7;80;Europe
Volkswagen Jetta;33.0;4;105.0;74.00;2190.;14.2;81;Europe
Volkswagen Model 111;27.0;4;97.00;60.00;1834.;19.0;71;Europe
Volkswagen Pickup;44.0;4;97.00;52.00;2130.;24.6;82;Europe
Volkswagen Rabbit;29.0;4;90.00;70.00;1937.;14.0;75;Europe
Volkswagen Rabbit;29.0;4;90.00;70.00;1937.;14.2;76;Europe
Volkswagen Rabbit;29.5;4;97.00;71.00;1825.;12.2;76;Europe
Volkswagen Rabbit;29.8;4;89.00;62.00;1845.;15.3;80;Europe
Volkswagen Rabbit;41.5;4;98.00;76.00;2144.;14.7;80;Europe
Volkswagen Rabbit C (Diesel);44.3;4;90.00;48.00;2085.;21.7;80;Europe
Volkswagen Rabbit Custom;29.0;4;97.00;78.00;1940.;14.5;77;Europe
Volkswagen Rabbit Custom;31.9;4;89.00;71.00;1925.;14.0;79;Europe
Volkswagen Rabbit Custom Diesel;43.1;4;90.00;48.00;1985.;21.5;78;Europe
Volkswagen Rabbit l;36.0;4;105.0;74.00;1980.;15.3;82;Europe
Volkswagen Scirocco;31.5;4;89.00;71.00;1990.;14.9;78;Europe
Volkswagen Super Beetle 117;0;4;97.00;48.00;1978.;20.0;71;Europe
Volkswagen Super Beetle;26.0;4;97.00;46.00;1950.;21.0;73;Europe
Volkswagen Type 3;23.0;4;97.00;54.00;2254.;23.5;72;Europe
Volvo 144ea;19.0;4;121.0;112.0;2868.;15.5;73;Europe
Volvo 145e (sw);18.0;4;121.0;112.0;2933.;14.5;72;Europe
Volvo 244DL;22.0;4;121.0;98.00;2945.;14.5;75;Europe
Volvo 245;20.0;4;130.0;102.0;3150.;15.7;76;Europe
Volvo 264gl;17.0;6;163.0;125.0;3140.;13.6;78;Europe
Volvo Diesel;30.7;6;145.0;76.00;3160.;19.6;81;Europe
ilovetrains28@cis106:~/Documents/sample_files/Csv$ sort -o sorted_cars.csv cars.csv
ilovetrains28@cis106:~/Documents/sample_files/Csv$ wc -l sorted_cars.csv
408 sorted_cars.csv
ilovetrains28@cis106:~/Documents/sample_files/Csv$ wc -m sorted_cars.csv
22663 sorted_cars.csv
ilovetrains28@cis106:~/Documents/sample_files/Csv$
```

Q2

challenge 1

```
ilovetrains28@cis106:~$ wc -l access.log
100 access.log
ilovetrains28@cis106:~$
```

Challenge 2

```
ilovetrains28@cis106:~$ head -1 access.log
2025-02-21 20:59:18 Touka /dashboard DENIED
```

Challenge 3

```
ilovetrains28@cis106:~$ tail -1 access.log
2025-03-04 03:30:08 Edward /logout FAIL
```

Challenge 4

```
ilovetrains28@cis106:~$ head -15 access.log
2025-02-21 20:59:18 Touka /dashboard DENIED
2025-11-15 18:52:10 Ken /dashboard DENIED
2025-04-22 04:04:31 Touka /logout FAIL
2025-03-08 05:42:35 Spike /admin SUCCESS
2025-02-07 16:08:25 Rize /download SUCCESS
2025-09-16 05:21:07 Ein /reset DENIED
2025-11-18 15:32:46 Touka /logout FAIL
2025-09-24 13:34:52 Julia /login SUCCESS
2025-02-17 04:17:29 Kei /reset FAIL
2025-05-05 03:36:25 Edward /404 SUCCESS
2025-12-27 08:24:08 Spike /404 FAIL
2025-11-24 03:27:58 Rei /404 DENIED
2025-01-02 16:03:31 Kei /settings DENIED
2025-03-22 23:09:42 Seras /reset FAIL
2025-12-02 08:57:33 Arisu /login SUCCESS
ilovetrains28@cis106:~$
```

Challenge 5

```
ilovetrains28@cis106:~$ tail -15 access.log
2025-02-23 15:01:58 Annie /reset SUCCESS
2025-11-01 10:13:13 Kei /login SUCCESS
2025-09-18 13:38:04 Honami /login DENIED
2025-04-22 17:48:13 Rei /download SUCCESS
2025-05-24 01:36:44 Faye /dashboard FAIL
2025-09-10 05:36:36 Eren /settings DENIED
2025-12-12 10:25:31 Haruka /settings DENIED
2025-01-09 09:22:09 Shinji /logout FAIL
2025-05-19 13:26:41 Honami /upload DENIED
2025-07-11 22:16:20 Asimov /login DENIED
2025-08-26 04:06:29 Ein /download DENIED
2025-01-13 07:08:53 Spike /dashboard SUCCESS
2025-08-08 10:49:33 Spike /settings FAIL
2025-06-16 19:25:20 Faye /logout DENIED
2025-03-04 03:30:08 Edward /logout FAIL
ilovetrains28@cis106:~$
```

Challenge 6

```
1:ilovetrains28@cis106: ~  
ilovetrains28@cis106:~$ cut -d ' ' -f3 access.log  
Touka  
Ken  
Touka  
Spike  
Rize  
Ein  
Touka  
Julia  
Kei  
Edward  
Spike  
Rei  
Kei  
Seras  
Arisu  
Rize  
Rei  
Kei  
Jet  
Seras  
Shinji  
Kei  
Annie  
Haruka  
Ken  
Touka  
Jet  
Eren  
Rize  
...
```

Challenge 7

```
1:ilovetrains28@cis106: ~  
2025-03-10 22:31:13 Shinji /logout FAIL  
2025-06-17 06:49:29 Shinji /upload DENIED  
2025-08-20 15:37:56 Spike /404 DENIED  
2025-12-27 08:24:08 Spike /404 FAIL  
2025-03-08 05:42:35 Spike /admin SUCCESS  
2025-06-21 16:22:08 Spike /admin SUCCESS  
2025-05-17 14:11:51 Spike /dashboard DENIED  
2025-10-14 16:27:59 Spike /dashboard DENIED  
2025-01-13 07:08:53 Spike /dashboard SUCCESS  
2025-09-15 18:17:31 Spike /download FAIL  
2025-04-19 13:45:21 Spike /reset FAIL  
2025-08-08 10:49:33 Spike /settings FAIL  
2025-06-24 12:10:18 Touka /404 FAIL  
2025-10-21 09:06:13 Touka /admin DENIED  
2025-02-21 20:59:18 Touka /dashboard DENIED  
2025-12-28 20:22:01 Touka /dashboard DENIED  
2025-11-23 20:57:19 Touka /download FAIL  
2025-04-22 04:04:31 Touka /logout FAIL  
2025-11-18 15:32:46 Touka /logout FAIL  
2025-11-18 01:24:34 Touka /settings SUCCESS  
2025-11-09 21:07:48 Walter /reset FAIL  
ilovetrains28@cis106:~$ sort -t ' ' -k 3 access.log -o sorted_access_log_by_username.log  
ilovetrains28@cis106:~$ ls  
access.log      essentials.sh    lab6-challenge1  music           project_jupiter  sorted_access_log_by_username.log  website_ch1_lab7_copy  
bash.pdf        File            lab6-challenge2  Music          project_venus    Templates         wildcardpractice  
cis106          hmw            main             Pictures        Public            Videos  
Desktop         inClassActivity man-page-activity.pdf portfolio        'Screenshot 2026-04-21 134341.png'  vscode.sh         website  
Documents       lab5files      marstenHouse    practicels      scripts          website           website_ch1_lab7  
Downloads       lab5files.zip  marstenhouse.sh practicels.tar.xz snap  
ilovetrains28@cis106:~$
```

Challenge 8

```
1:ilovetrains28@cis106: ~  
cis106          hmw            main             Pictures        Public            Videos  
Desktop         inClassActivity man-page-activity.pdf portfolio        'Screenshot 2026-04-21 134341.png'  vscode.sh         website  
Documents       lab5files      marstenHouse    practicels      scripts          website           website_ch1_lab7  
Downloads       lab5files.zip  marstenhouse.sh practicels.tar.xz snap  
ilovetrains28@cis106:~$ sort -t ' ' -k 4 access.log -o sorted_access_log_by_endpoint.log  
ilovetrains28@cis106:~$ ls  
access.log      essentials.sh    lab6-challenge1  music           project_jupiter  sorted_access_log_by_endpoint.log  website_ch1_lab7  
bash.pdf        File            lab6-challenge2  Music          project_venus    sorted_access_log_by_username.log  website_ch1_lab7_copy  
cis106          hmw            main             Pictures        Public            Templates         wildcardpractice  
Desktop         inClassActivity man-page-activity.pdf portfolio        'Screenshot 2026-04-21 134341.png'  vscode.sh         website  
Documents       lab5files      marstenHouse    practicels      scripts          website           website_ch1_lab7  
Downloads       lab5files.zip  marstenhouse.sh practicels.tar.xz snap  
ilovetrains28@cis106:~$ sort -t ' ' -k 4 access.log  
2025-08-20 15:37:56 Spike /404 DENIED  
2025-11-24 03:27:58 Rei /404 DENIED  
2025-06-24 12:10:18 Touka /404 FAIL  
2025-12-27 08:24:08 Spike /404 FAIL  
2025-02-22 08:48:06 Seras /404 SUCCESS  
2025-05-05 03:36:25 Edward /404 SUCCESS  
2025-03-22 21:31:43 Asimov /admin DENIED  
2025-04-24 21:30:36 Honami /admin DENIED  
2025-10-21 09:06:13 Touka /admin DENIED  
2025-06-25 15:49:35 Honami /admin FAIL  
2025-01-27 22:30:42 Eren /admin SUCCESS  
2025-03-08 05:42:35 Spike /admin SUCCESS  
2025-06-21 16:22:08 Spike /admin SUCCESS  
2025-11-22 07:40:35 Edward /admin SUCCESS  
2025-01-17 08:01:57 Rei /dashboard DENIED  
2025-01-25 22:28:42 Asimov /dashboard DENIED  
2025-02-21 20:59:18 Touka /dashboard DENIED  
2025-05-13 11:11:51 Spike /dashboard DENIED
```

Challenge 9

```
1: ilovetrains28@cis106: ~  
ilovetrains28@cis106:~$ cut -d ' ' -f3,4 --output-delimiter=' accessed: ' access.log  
Touka accessed: /dashboard  
Ken accessed: /dashboard  
Touka accessed: /logout  
Spike accessed: /admin  
Rize accessed: /download  
Ein accessed: /reset  
Touka accessed: /logout  
Julia accessed: /login  
Kei accessed: /reset  
Edward accessed: /404  
Spike accessed: /404  
Rei accessed: /404  
Kei accessed: /settings  
Seras accessed: /reset  
Arisu accessed: /login  
Rize accessed: /logout  
Rei accessed: /login  
Kei accessed: /reset  
Jet accessed: /reset  
Seras accessed: /404  
Shinji accessed: /upload  
Kei accessed: /download  
Annie accessed: /login  
Haruka accessed: /reset  
Ken accessed: /login  
Touka accessed: /settings  
Jet accessed: /upload  
Eren accessed: /logout  
Rize accessed: /upload
```